

Parth Wadhwa

IIT Delhi | B.Tech Chemical Engineering | AI Full-Stack Developer

parthwadhwa1980@gmail.com | +91-9057669333

github.com/ParthWadhwa14 | https://www.linkedin.com/in/parth-wadhwa-855650323/

Education

Indian Institute of Technology (IIT) Delhi

B.Tech in Chemical Engineering

CGPA: 8.05/10

New Delhi, India

2024 – Present

Experience

AI Intern – Fumind.ai

Remote

Feb – April 2026

- Built a one-shot image segmentation pipeline using OpenCV and vision-language models to segment objects from a single reference image.
- Worked on AI-assisted computer vision workflows involving preprocessing, contour extraction, and prompt-guided segmentation.
- Integrated the system into the company monorepo while ensuring compatibility with existing deployment pipelines.
- Collaborated in a fast-paced development environment using Git-based workflows and AI coding tools.

Projects

Research Agentic Chatbot

LangGraph, FastAPI, Next.js, Python, Firebase

- Engineered a full-stack AI research assistant utilizing LangGraph and FastAPI to execute complex, multi-step agentic workflows for web search synthesis, document-grounded retrieval, and persistent contextual memory.
- Implemented multimodal processing pipelines for PDF text extraction and image understanding, alongside automated local data analysis using pandas and matplotlib to generate statistical insights and charts.
- Designed a responsive Next.js frontend with Firebase authentication and Firestore state management, featuring real-time SSE streaming and dynamic generation of downloadable PPTX, LaTeX, and Markdown artifacts.

Multimodal Visual Search Engine

YOLOv11, CLIP, FAISS, PyTorch, Streamlit

- Engineered a production-inspired multimodal visual search engine utilizing YOLOv11 for object detection metadata and CLIP for generating high-dimensional semantic image and text embeddings.
- Implemented a hybrid retrieval pipeline using a FAISS vector database, enabling text-to-image, image-to-image, and object-filtered queries scored via a custom weighted ranking algorithm.

ResearchSync – AI-Powered Research Collaboration Platform

React, Node.js, Supabase, Gemini API

- Built a full-stack collaborative platform with real-time chat, Kanban task management, and shared research workspaces.
- Implemented authentication, encrypted storage, and team-based access control using Supabase Auth

SehatSaathi-AI

React, Node.js, AI APIs

- Built an AI-powered healthcare assistance platform providing symptom-based guidance, medical information, and intelligent conversational support.
- Integrated LLM-based workflows and AI APIs to deliver context-aware health recommendations and improve user interaction experience.

Mental-Health-AI-Copilot

React, Node.js, Generative AI

- Developed an AI-driven mental health support assistant capable of empathetic conversations, personalized guidance, and wellness-focused interactions.
- Integrated large language models for contextual response generation, emotional support prompts, and workflows.

Technical Skills

Languages: Python, JavaScript, C++, Java, SQL, GO

Agentic Frameworks & Automation: LangGraph, LangChain, CrewAI, AutoGen, MCP, n8n

Frameworks & Libraries: React, Node.js, TensorFlow, PyTorch, scikit-learn, OpenCV

AI & LLMs: OpenAI API / SDK, Gemini API, Hugging Face, LLaMA, Prompt Engineering

Tools: Git, GitHub, Jupyter Notebook, Figma

Core Areas: Agentic Workflows, Full-Stack Development, Computer Vision, Machine Learning

Achievements

- JEE Advanced 2024 AIR 1980 among 180,000+ candidates.
- JEE Main 2024 AIR 1500 among 1.4 million+ candidates.
- MLL100: Awarded first position for making the best assignment.

Leadership & Activities

- Technical Coordinator – BnC, IITD
- Vice President, 2D Society – Design Club, IIT Delhi
- Design Coordinator – BSW, IITD